Mathematical Foundation of Computer Science

QUIZ – Course Pre-requisite

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1. What is the number of integral solutions of the equation: x1+x2 = 10 with x1>=3 and x2>=1?
   A. 9C6 B. 9C4 C. 10C4 D. 5C2

2. S be a set of ordered pairs (a, b) of objects, where the first object a comes from a set of size p, and for each choice of object a there are q choices for object b. What is the size of S?
   A. pxq B. p+q C. p^q D. px(q)!

3. Let A and B be two arrays of size n, each containing the numbers {1, 2, . . . , n} in some order. The operations allowed on the entries of the arrays are comparisons between two entries and swapping two entries. Consider the best possible algorithm to transform the array A to array B using the above two operations only. What is the total number of operations in which the above algorithm can perform this task in the worst case?
   A. n B. n^2 C. n! D. Constant
4. Let A1 and A2 be two arbitrarily chosen subsets of {1, 2, . . . , n}. The number of ways such that A1 ∩ A2 = ∅ is
   A. 2^n B. 3^n C. 3^{n-1} D. 4^n-3^n
5. Which of the following(s) are Always true?
   A. A tree is a acyclic connected graph
   B. A n-vertex tree has n-1 edges
   C. The vertices of a tree can't be colored using 2 colours such that no adjacent vertices receive same colours.
   D. A tree is a planar graph
6. If P and Q are propositions, then the expression (P ∧ (P → Q)) → Q is
   A. Always true
   B. Always false
   C. True for only two assignments
   D. False for only one assignment
7. There are two coins in a box. One of the coins is a fair coin and for the other, the probability of head is 2/3. A coin is chosen at random from the box and tossed 2 times. What is the probability that both toss results a head?
   A. 1/3 B. 4/9 C. 25/72 D. 25/36

8. Let T(n) be the count of n-digit numbers (using digits 0, 1, . . . , 9, with first digit non-zero) such that no three consecutive digits are the same. Then which of the following is a correct recurrence relation for T(n)? Note that T(1) = 9 and T(2) = 90.
   A. T(n)=9T(n-1) + 9T(n-2)
   B. T(n)=9T(n-1) + 8T(n-2) + 9
   C. T(n)=9T(n-1) + 9
   D. T(n)=9T(n-1) 9T(n-2)

9. Let G = (V, E) be an undirected simple graph, and s be a designated vertex in G. For each $v \in V$, let d(v) be the length of a shortest path between s and v. For an edge (u, v) in G, what can not be the value of d(u) − d(v)?
   A. 1 B. 0 C. -1 D. 2

10. Suppose a machine generates an integer uniformly at random from {0, 1, . . . , n} and whenever it is less than an integer k it prints "OK". If it prints "OK" with a probability 0.5, what is the guaranteed relation between n and k?
   (A) k = n/2
   (B) k = (n − 1)/2
   (C) k = (n + 1)/2
   (D) Nothing can be inferred

11. Let the variables x and y vary over the set of natural numbers, N. Consider the formulas expressing properties about N: (i) $\forall x \exists y (x < y)$, and (ii) $\exists y \forall x (x \leq y)$. Which of the following statements is correct?
   (A) Both (i) and (ii) hold.
   (B) (i) holds but (ii) does not hold.
   (C) (ii) holds but (i) does not hold.
   (D) Both (i) and (ii) do not hold.

12. Find the value of 128^{129} mod 17
   A. 9 B. 7 C. 2 D. 5

13. Solution to the equation $x^{103} \equiv 4 \bmod 11$ is
   A. $x \equiv 5 \bmod 11$ B. $x \equiv 3 \bmod 11$ C. $x \equiv 4 \bmod 11$ D. $x \equiv 7 \bmod 11$

14. If a(t), b(t), and c(t) are the lengths of the three sides of a triangle t in non-decreasing order (i.e. $a(t) \leq b(t) \leq c(t)$), we define three sets as follows:
   X: = {Triangle t : a(t) = b(t)},
   Y := {Triangle t : b(t) = c(t)}
   T := the set of all triangles
   Which of the following(s) represents the set of all equilateral triangles (all sides equal)

   A. $X \cap Y$
   B. X U Y
   C. X-Y
   D. None of these

15. Let A1, A2, A3, and A4 be four matrices of dimensions 8 x 5, 5 x 10, 10 x 8, and 8 x 5, respectively. The minimum number of scalar multiplications required to find the product A1A2A3A4 using the basic matrix multiplication method is
   (A) 800 (B) 850 (C) 920 (D) 1360